Cellular Coverage & Signal Strength Data Pre-processing & Upload Guidelines (SOP)

UNDER ACTIVE DEVELOPMENT / WORK-IN-PROGRESS: The **Cellular Coverage** import module and database pipeline are currently **under active development in an ongoing branch**. Template specifications, schema rules, and ingestion behaviors represent work-in-progress standards and are subject to change upon final release. Coordinate with the project engineering team before uploading production coverage datasets.

Strict Pre-Processing Rule: All files **must** be sanitized and realigned based on the **latest official Excel template**. The system will only accept post-processed files. Any unformatted, misaligned, or missing required fields will cause rows to fail or the batch to abort.

1. Summary Checklist for Uploaders

Before uploading any Cellular Coverage file, verify that:

- Columns match the official template exactly: `ServiceProvider`, `Network`, `Lat`, `Lng`, `Strength`, `Address`.
- Data rows begin on **Row 2** immediately below the single header row.
- Required fields (`ServiceProvider`, `Network`, `Lat`, `Lng`, `Strength`) are 100% populated.
- `ServiceProvider` matches one of the registered **Wireless Service Providers** in the system.
- `Network` matches an exact pre-configured network name in the QoS portal (e.g., `DITO 4G 2600`, `GLOBE 4G 1800`, `SMART 5G 3500`).
- `Strength` is a **clean numeric value** with the negative sign preserved (e.g., `-85.5`, not `"-85.5 dBm"`).
- Coordinate fields (`Lat`, `Lng`) contain clean WGS84 decimals without degree symbols or directional letters (`N`, `E`).
- Latitude and Longitude values are not swapped (`Lat` in Col C, `Lng` in Col D).
- Optional field `Address` is formatted cleanly; empty cells are left **BLANK** (no `"---"`, `"N/A"`, or `"null"`).

2. Required Fields (Mandatory Columns)

The following 5 columns are **Mandatory Required Fields**. If any cell is blank or invalid, that row cannot be imported:

Header Name	Col	Type	Description	Action Required if Missing / Invalid
`ServiceProvider`	A	Dropdown	Registered wireless provider	Select from approved dropdown list
`Network`	B	Dropdown	Network technology and band configuration	Select exact network name pre-configured in portal
`Lat`	C	Decimal	WGS84 Latitude coordinate	Ensure valid decimal format; delete row if unavailable
`Lng`	D	Decimal	WGS84 Longitude coordinate	Ensure valid decimal format; delete row if unavailable

Header Name	Col	Type	Description	Action Required if Missing / Invalid
`Strength`	E	Decimal	Measured RF signal strength (dBm)	Enter clean decimal value preserving negative sign; delete row if empty
`Address`	F	Text	Optional location or descriptive address	Optional; leave blank if unavailable

> **Caution:** If a row is missing `ServiceProvider`, `Network`, `Lat`, `Lng`, or `Strength`, **delete the entire row** from the spreadsheet before uploading.

3. Signal Strength (Strength) Sanitization

The `Strength` column represents RF field strength measurements (typically measured as RSRP or RSSI in dBm).

- **Preserve the Negative Sign (-):** Mobile RF signal measurements are almost always negative (e.g., `-70` to `-120 dBm`). Do not omit the minus symbol.
- **No Units:** Strip all text units such as `dBm`, `dB`, or `m` (e.g., `-88.5 dBm` → `-88.5`).
- **No Commas or Spaces:** Replace decimal commas with periods (e.g., `-92,4` → `-92.4`).
- **Valid Signal Range:** Values typically range from `-140.0` (extremely weak) to `-44.0` (extremely strong).

4. Dropdown Columns Validation

Column	Allowed Values / Standard	Action Required
`ServiceProvider`	Registered Wireless Operators (e.g., `DITO TELECOMMUNITY CORP.`, `GLOBE COMMUNICATIONS INC.`, `SMART COMMUNICATIONS INC.`)	Select from template dropdown list. Unrecognized operators will cause rows to fail.
`Network`	Registered Network Names (e.g., `DITO 4G 2600`, `GLOBE 4G 1800`, `SMART 5G 3500`)	Must match an existing Network configured in the database under that Provider. If a new network standard is introduced, request admin setup prior to upload.

> **Network-Provider Alignment:** The selected `Network` must belong to the specified `ServiceProvider`. If a row lists `GLOBE` with a `DITO 4G 2600` network, the system will not be able to link the coverage point.

5. Spatial Deduplication & Checkpoint Resume

Feature	Behavior
Spatial Deduplication	The system creates spatial indexes based on `(NetworkId, BandId, Location)`. Repeated identical coordinates for the same network are deduplicated to optimize map rendering performance.
Checkpoint Resume	In the event of large multi-megabyte datasets, the system saves progress checkpoints periodically. Interrupted uploads can resume from the last saved row automatically.

6. Post-Upload Support

If your file encounters processing errors:

1. Check the upload status dialog in the **Cellular Coverage** tab.
2. Confirm that selected `Network` names exist in the system directory.
3. Review any reported rows for malformed coordinates or non-numeric strength entries.
4. Contact the system administrator with the error message for further assistance.